

Mobile Payment

B2W – W2B service

Bank to Wallet – Wallet to Bank

Mobile operator ⇔ Bank Interface
Specifications

Version 3.1

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1 Service's overview

The service aims at providing customers with means to remotely exchange money between their mobile money wallet and their bank account.

The Bank provides the web services dedicated to transaction, account balance and mini-statement processing.

The Mobile Operator sends requests to process these operations.

2 List of Functionalities

functionalities	Description
AccountToWalletTransfer	to transfer money from a bank account to a mobile account
WalletToAccountTransfer	to transfer money from a mobile account to a bank account
TransferStatusInquiry	to check the status of a transfer
CancelTransfer	to cancel a AccountToWalletTransfer or WalletToAccountTransfer operation
GetAccountBalance	to get the balance of a bank account
GetMiniStatement	to get the mini-statement of a bank account

3 Description of exchanged parameters

Here is the list of parameters that are exchanged (in requests or/and in responses), with their descriptions and some samples.

Field Name	Data Type	Length	Description	Streams
operatorCode	string	20	unique code identifying the the mobile operator. e.g ORANGEMONEYML	all requests, all responses
requestId	string	16	request's identifier. Alphanumeric, and must be unique per request. e.g OR150123ITGA5Y : 1-2 - Two-letter unique code of Mobile Money Operator 3-4 - YY (Year) 5-6 - MM (Month) 7-8 - DD (Day) 9-16 - Random alphanumeric	all requests, all responses
requestToken	string	128	SHA 512 Hashing of [Account Alias + requestId + operatorCode + Password + Amount. e.g.: Alias: BICBIDON000000000000001 Request ID: OR151029ABCDEFGH Operator Code: ORANGEMONEYML Password: foobar Amount: 1565 →SHA512(BICBIDON000000000000001OR151029ABCDEFGHORANGEMONEYMLfoobar1565)= 8242765191be4064f4a5d1ea38c831d38b5b79a5311b365f031b95c81d26935096be6d7236572b036d142d6bfe1c1c6056880dcde90d7accd591b61b22cd9bd8	AccountToWalletTransfer, WalletToAccountTransfer
requestType	string	20	type of the request: A2W (bank Account To mobile Wallet transfer), W2A (mobile Wallet To bank Account transfer), GETACCBAL (bank Account Balance inquiry) TRANINQ (Transaction status Inquiry) GETMINI (Mini statement of bank account request) CANCELTRAN (Transaction Cancelation request)	all requests
affiliateCode	string	3	ry affiliate code of the bank. e.g.: OML : Mali, OMG : Madagascar, OCI : Ivory Cost	all requests, all responses
responseCode	string	3	CF § list of response codes. e.g.: 000 (<=> Success)+F31, other = failure codes	all responses
responseMessage	string	50	response message. CF § list of response codes. e.g.: Success	all responses
tranRefNo	string	16	transaction reference.	GetMiniStatementResponse

tranDate	date	24	transaction date, with 8601 ISO format = - YYYY-MM-DD for AccountToWalletTransfer and WalletToAccountTransfer (e.g.: 2015-10-23) - YYYY-MM-DDThh:mm:ss.sssZ for GetMiniStatementResponse (e.g.: 2015-10-23T10:48:33.000Z)	AccountToWalletTransfer, WalletToAccountTransfer, GetMiniStatementResponse
tranType	string	3	transaction type. Not used (empty field)	GetMiniStatementResponse
ccy	string	3	local currency in 4217 ISO Code. e.g.: XOF (Mali, Sénégal,...), MGA (Madagascar), BWP (Botswana)	AccountToWalletTransfer, WalletToAccountTransfer, GetAccountBalanceResponse, GetMiniStatementResponse
crDr	string	1	transaction type: C - Credit (mobile wallet to bank account), D-Debit (bank account to mobile wallet)	GetMiniStatementResponse
amount	double		transaction amount. e.g.: 200.15	AccountToWalletTransfer, WalletToAccountTransfer, GetMiniStatementResponse
narration	string	105	transaction details. Not used (empty field)	GetMiniStatementResponse
externalRefNo	string	16	reference of a transaction: - either same as requestId (for WalletToAccountTransfer, AccountToWalletTransferResponse), - or banking reference of original transaction (for CancelTransfer, TransferStatusInquiry)	AccountToWalletTransfer, AccountToWalletTransferResponse, WalletToAccountTransfer, WalletToAccountTransferResponse, TransferStatusInquiry, TransferStatusInquiryResponse, CancelTransfer, CancelTransferResponse
CBAReferenceNo	string	16	banking reference of original transaction	AccountToWalletTransferResponse, WalletToAccountTransferResponse, TransferStatusInquiryResponse, CancelTransferResponse
mobileNo	string	20	MSISDN of wallet. Not used (empty field)	AccountToWalletTransfer, WalletToAccountTransfer
mobileName	string	50	name of customer for wallet . Not used (empty field)	AccountToWalletTransfer, WalletToAccountTransfer
mobileAlias	string	22	alias for wallet. Not used (empty field)	AccountToWalletTransfer, WalletToAccountTransfer
accountNo	string	20	bank account No. Not used (empty field)	AccountToWalletTransfer, WalletToAccountTransfer, GetAccountBalance, GetAccountBalanceResponse, GetMiniStatement
accountAlias	string	22	user identifier. e.g.: BICBICBI0000000001300	AccountToWalletTransfer, WalletToAccountTransfer, GetAccountBalance, GetAccountBalanceResponse, GetMiniStatementResponse
accountName	string	50	name for bank account. Not used (empty field)	AccountToWalletTransfer, WalletToAccountTransfer, GetAccountBalanceResponse
transferDescription	string	50	description of the transaction. Not used (empty field)	AccountToWalletTransfer, WalletToAccountTransfer
charge	double		charge of the transfer. Not used, default value: 0	AccountToWalletTransfer, WalletToAccountTransfer
reason	string	50	transaction details. Not used (empty field)	GetAccountBalance, GetMiniStatementResponse
branchCode	string	3	bank branch code of account. Not used (empty field)	GetAccountBalanceResponse
availableBalance	double		user available balance in the bank account. Not used	GetAccountBalanceResponse
currentBalance	double		user current balance in the bank account	GetAccountBalanceResponse
accountType	string	1	bank account. Not used (empty field)	GetAccountBalanceResponse
accountClass	string	1	Internal to Core Banking Application. Not used (empty field)	GetAccountBalanceResponse
accountStatus	string	10	account status. Not used (empty field)	GetAccountBalanceResponse
udf1, udf2, udf3	string	50	User Defined Fields, for future use. (Empty fields)	AccountToWalletTransfer, WalletToAccountTransfer, GetAccountBalanceResponse

4 AccountToWalletTransfer: bank account to mobile account transfer

to transfer money from a bank account to a mobile account

Operation	AccountToWalletTransfer			
Request (AccountToWalletTransfer) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier	OR151023OSR023
requestToken	string	128	token	ea9ce3f18ecc96bbfeb63381689b9497e1b9f30fd00d80560d5db70bc1229d4d4ba9fb80935fa4d3836755fdbd72800dc0f0b04ada7e09672dcb289346d4fd47
requestType	string	20	A2W (bank Account To mobile Wallet transfer)	A2W
affiliateCode	string	3	country affiliate code of the bank.	EML
externalRefNo	string	16	reference of a transaction: same as requestId	OR151023OSR023
mobileNo	string	20	Not used (empty field)	empty
mobileName	string	50	Not used (empty field)	empty
mobileAlias	string	22	Not used (empty field)	empty
accountNo	string	20	Not used (empty field)	empty
accountAlias	string	22	user identifier	BICBICBI00000000001300
accountName	string	50	Not used (empty field)	empty
transferDescription	string	50	Not used (empty field)	empty
ccy	string	3	local currency	XOF
amount	double		transaction amount	1000
charge	double		Not used, default value: 0	0
tranDate	date	24	transaction date	2015-10-23
udf1	string	50	Not used (empty field)	empty
udf2	string	50	Not used (empty field)	empty
udf3	string	50	Not used (empty field)	empty
Response (AccountToWalletTransferResponse) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier (same as request!)	OR151023OSR023
affiliateCode	string	3	country affiliate code of the bank.	EML
responseCode	string	3	CF list of response codes.	000
responseMessage	string	50	CF list of response codes.	Success
externalRefNo	string	16	reference of a transaction: same as requestId. Not used	OR151023OSR023
CBARreferenceNo	string	16	banking reference of transaction	FKGI22541

5 WalletToAccountTransfer: mobile account to bank account transfer

to transfer money from a mobile account to a bank account

Operation	WalletToAccountTransfer			
Request (WalletToAccountTransfer) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	GEMONEYML
requestId	string	16	request's identifier	1023T4783D
requestToken	string	128	token	f18ecc96bbfeb63381689b9497e1b9f30fd00 d80560d5db70bc1229d4d4ba9fb80935fa4d383675 5fdbd72800dc0f0b04ada7e09672dcb289346d4fd47
requestType	string	20	W2A (mobile Wallet To bank Account transfer)	
affiliateCode	string	3	country affiliate code of the bank.	
externalRefNo	string	16	reference of a transaction: same as requestId	1023T4783D
mobileNo	string	20	Not used (empty field)	empty
mobileName	string	50	Not used (empty field)	empty
mobileAlias	string	22	Not used (empty field)	empty
accountNo	string	20	Not used (empty field)	empty
accountAlias	string	22	user identifier	CBI0000000001300
accountName	string	50	Not used (empty field)	empty
transferDescription	string	50	Not used (empty field)	empty
ccy	string	3	local currency	
amount	double		transaction amount	1000
charge	double		Not used, default value: 0	0
tranDate	date	24	transaction date	2015-10-23
udf1	string	50	Not used (empty field)	empty
udf2	string	50	Not used (empty field)	empty
udf3	string	50	Not used (empty field)	empty
Response (WalletToAccountTransferResponse) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	GEMONEYML
requestId	string	16	request's identifier (same as request!)	1023T4783D
affiliateCode	string	3	country affiliate code of the bank.	EML
responseCode	string	3	CF list of response codes.	000
responseMessage	string	50	CF list of response codes.	Success
externalRefNo	string	16	reference of a transaction: same as requestId. Not used	1023T4783D
CBAResponseNo	string	16	banking reference of transaction	2541

6 TransferStatusInquiry: transfer's status check

to check the status of a transfer.

in response :

- CBAReferenceNo filled <=> original transaction OK on bank side
- CBAReferenceNo empty <=> original transaction not found or not valid on bank side (i.e. cancellation has to be done on mobile operator side)

Operation		TransferStatusInquiry			
Request (TransferStatusInquiry) Parameters					
Field Name	Data Type	Length	Description	Sample	
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML	
requestId	string	16	request's identifier	OR150313C9OL8X	
requestToken	string	128	Not used (empty field)	empty	
requestType	string	20	TRANINQ (Transaction status Inquiry)	TRANINQ	
affiliateCode	string	3	country affiliate code of the bank.	EML	
externalRefNo	string	16	reference (requestId) of original transaction which has to be checked	OR151023OSR023	
Response (TransferStatusInquiryResponse) Parameters					
Field Name	Data Type	Length	Description	Sample	
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML	
requestId	string	16	request's identifier (same as request!)	OR150313C9OL8X	
affiliateCode	string	3	country affiliate code of the bank.	EML	
responseCode	string	3	CF list of response codes.	000	
responseMessage	string	50	CF list of response codes.	Success	
externalRefNo	string	16	reference of a transaction: same as requestId. Not used	OR150313C9OL8X	
CBAReferenceNo	string	16	original transaction reference if found on bank side, empty if transaction not found	FKGI22541	

7 CancelTransfer: transfer cancellation

to cancel a AccountToWalletTransfer or WalletToAccountTransfer operation

Operation	CancelTransfer			
request (CancelTransfer) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier	OR151023ITGA5Y
requestToken	string	128	Not used (empty field)	empty
requestType	string	20	CANCELTRAN (Transaction Cancellation request)	CANCELTRAN
affiliateCode	string	3	country affiliate code of the bank.	EML
externalRefNo	string	16	banking reference (CBAResponseNo) of original transaction which has to be cancelled	FKGI22541
Response (CancelTransferResponse) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier (same as request!)	OR151023ITGA5Y
affiliateCode	string	3	country affiliate code of the bank.	EML
responseCode	string	3	CF list of response codes.	000
responseMessage	string	50	CF list of response codes.	Success
externalRefNo	string	16	reference of a transaction: same as requestId. Not used	OR151023ITGA5Y
CBAResponseNo	string	16	banking reference of this cancellation	FKGI25894

8 GetAccountBalance: bank account balance

to get the balance of a bank account

Operation	GetAccountBalance			
Request (GetAccountBalance) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier	OR151023UKTR4W
requestToken	string	128	Not used (empty field)	empty
requestType	string	20	GETACCBAL (bank Account Balance inquiry)	GETACCBAL
affiliateCode	string	3	country affiliate code of the bank.	EML
accountNo	string	20	Not used (empty field)	empty
accountAlias	string	22	user identifier	CBI0000000001300
reason	string	50	Not used (empty field)	empty
Response (GetAccountBalanceResponse) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier (same as request!)	OR151023UKTR4W
affiliateCode	string	3	country affiliate code of the bank.	EML
responseCode	string	3	CF list of response codes.	000
responseMessage	string	50	CF list of response codes.	Success
accountNo	string	20	Not used (empty field)	empty
accountAlias	string	22	user identifier	CBI0000000001300
accountName	string	50	Not used (empty field)	empty
ccy	string	3	local currency	XOF
branchCode	string	3	Not used (empty field)	empty
availableBalance	double		user's bank current balance. Not used	1450
currentBalance	double		user's bank available balance	1450
accountType	string	1	Not used (empty field)	empty
accountClass	string	1	Not used (empty field)	empty
accountStatus	string	10	Not used (empty field)	empty
udf1	string	50	Not used (empty field)	empty
udf2	string	50	Not used (empty field)	empty
udf3	string	50	Not used (empty field)	empty

9 GetMiniStatement: bank account mini-statement

to get the mini-statement of a bank account

Operation	GetMiniStatement			
Request (GetMiniStatement) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier	1023BD6KG0
requestToken	string	128	Not used (empty field)	empty
requestType	string	20	GETMINI (Mini statement of bank account request)	GETMINI
affiliateCode	string	3	country affiliate code of the bank.	EML
accountNo	string	20	Not used (empty field)	empty
accountAlias	string	22	user identifier	CBI0000000001300
reason	string	50	Not used (empty field)	empty
Response (GetMiniStatementResponse) Parameters				
Field Name	Data Type	Length	Description	Sample
operatorCode	string	20	mobile operator identifier	ORANGEMONEYML
requestId	string	16	request's identifier (same as request!)	OR151023BD6KG0
affiliateCode	string	3	country affiliate code of the bank.	EML
responseCode	string	3	CF list of response codes.	000
responseMessage	string	50	CF list of response codes.	Success
List of Transaction				
tranRefNo	string	16	transaction reference.	FKGI22541
tranDate	dateTime	24	transaction date. YYYY-MM-DDThh:mm:ss.sssZ	2015-10-23T10:48:33.000Z
tranType	string	3	Not used (empty field)	empty
ccy	string	3	local currency	XOF
crDr	string	1	transaction type: C - Credit (W2B) , D-Debit (B2W)	D
amount	double		transaction amount	5784
narration	string	105	Not used (empty field)	empty

10 Return codes

Here is the list of returned codes scanned by the mobile operator.

A few of them are treated by the mobile operator to inform customers or to reverse a transfer.

List of Error Codes	
Code	Description
000	Success
010	Bank Technical Error
E01	Bank Technical Error
E02	Bank Technical Error
E03	Bank Technical Error
E04	Bank Technical Error
E05	Bank Technical Error
E06	Bank Technical Error
E07	Bank Technical Error
E08	Bank Technical Error
E09	Bank Technical Error
E10	Bank Technical Error
E11	Operation in Timeout. Use TransferStatusInquiry to check status
E12	Bank Technical Error
E13	Unauthorized Request on Account (crossed thresholds)
E14	Unauthorized Request on Account (crossed thresholds)
E15	Unauthorized Request on Account (crossed thresholds)
E16	Insufficient Balance
E17	Bank Technical Error
E18	Bank Technical Error
E19	Bank Technical Error
E20	Bank Technical Error
E21	Bank Technical Error
E22	Transaction Violates Account's Minimum Balance
E35	Bank Technical Error
E99	Bank Technical Error